

2 Git

Git is a free tool that can handle both small and large projects with great efficiency. It is an open source distributed version control system that offers flexibility for modifying the source code as per the user's requirements. It has a branching model, which helps every sysadmin to create multiple local branches that are independent as well. Git has various features such as multiple workflows, colour controls, easier staging areas, good GPG signing for tags, etc. This amazing version control system offers functionalities free, unlike other systems that give paid functionalities to the user.

Features

- It supports branches.
- It is easy to track changes with Git.
- Git can fix issues easily.
- Collaboration becomes simpler due to the shared repository.

3 Bootstrap

Bootstrap was formerly known as Twitter Blueprint, and was developed by Twitter as the framework for increasing the reliability of open source toolkits. Users can design and develop CSS, JavaScript and HTML applications through Bootstrap. It is an open source framework for sysadmins that offers free facilities to the user.

Features

- Bootstrap offers button components and custom cards.
- It focuses on the utility classes for minimising CSS bloat.
- It has custom utility classes of extended functionality.

4 Kubernetes

Designed and developed by Google, Kubernetes is a powerful system that offers horizontal scaling to sysadmins. They can scale an application with the help of a single command (through the user interface), according to the CPU usage. Users can install Kubernetes with any package manager, such as *apt-get* or *yum*. This open source tool has different servers called nodes that help in configuring container networking—each of these nodes contacts a cluster group by a kubelet. Kubernetes is open source and can move data on-premise, as well as to hybrid and public cloud infrastructure.

Features

- It offers automated scheduling.
- Kubernetes has a self-healing capability.
- Users can use automated rollouts for the process.
- For development, production and testing, users get environment consistency.

5 SystemRescueCD

This open source tool for Windows sysadmins is a Linux based package, which is used to troubleshoot the system of Windows and Linux. You can download the SystemRescueCD ISO file so that you can use it for creating a bootable USB drive. It supports various file systems like *btrfs*, *xfs*, *ext2/ext3/ext4*, *vfat*, *ntfs*, *reiserfs*, *nfs* and *samba*. SystemRescueCD offers file editing, bootloader restoration, and network troubleshooting tools.

Features

- It has an *fdisk* feature for editing the disk partition tables.
- It offers GParted and GNU Parted abilities for portioning disks and resizing these partitions.
- There is a *ddrescue* feature for extracting recoverable data.
- SystemRescueCD provides archiving and unarchiving abilities.

6 Docker

Docker is one of the best open source tools for Windows sysadmins because applications can be developed and run in virtual containers. There are two different versions of Docker — Docker Community Edition (DCE) and Docker Enterprise Edition (DEE). DCE provides tools for building applications, and DEE provides multi-architecture operations for IT sectors. Docker has low overheads and a small footprint that offers flexibility.

Features

- Docker is easy and offers faster configuration.
- It can help to improve productivity.
- It provides application isolation.
- There is great security management.

7 Clonezilla

Clonezilla is an open source tool for Windows sysadmins, for disk cloning and imaging. It supports various file systems such as *ext2*, *ext3*, *ReiserFS*, *xfs*, and *jfs*. Apart from that, Clonezilla also supports the FAT/NTFS files on the Windows system for the cloning operating systems of GNU/Linux and Windows. This tool uses multicast for cloning multiple disks. There are two versions of Clonezilla — Clonezilla Live and Clonezilla SE.

Features

- Clonezilla offers multi-file system support.
- It offers customisable imaging and cloning.
- Users can do image encryption.
- There is AES-256 encryption.